

# MachinelQ

## ERP Product Specification & Release Plan

Project-based machine manufacturing - from customer enquiry to lifetime service

VERSION 1.0

07 September 2026

Purpose: Client scope alignment,  
product development and release  
acceptance

DEVELOPED BY  
DEVELOPED FOR

**Quorin Tech**  
**MacProAutomation**

# 0. Document Control

| Field            | Definition                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------|
| Document         | MachinelQ ERP Product Specification & Release Plan                                                    |
| Version          | 1.0                                                                                                   |
| Status           | Baseline for client review and controlled development                                                 |
| Developed by     | Quorin Tech                                                                                           |
| Developed for    | MacProAutomation                                                                                      |
| Source scope     | Original client Dashboard document plus MachinelQ OEM workflow corrections                            |
| Primary audience | Client stakeholders, product owner, engineering, development, QA and implementation teams             |
| Change control   | Any change to scope, workflow, acceptance criteria or release boundary must be recorded and approved. |

## Contents

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# 1. Purpose, Product Principles and Scope

This specification defines what MachineIQ shall deliver, how the modules connect, and the minimum conditions for accepting each release.

PRODUCT ANCHOR

**The Machine Project is the central business object. Sales, engineering, BOM, procurement, inventory, production, quality, delivery, finance and service records must remain traceable to it.**

## 1.1 Product objective

- Provide a single operational system for project-based OEM and special-purpose machine manufacturers.
- Control the full lifecycle from enquiry and quotation through engineering, procurement, assembly, FAT, commissioning, warranty and service.
- Replace disconnected spreadsheets and informal follow-up with accountable records, statuses, approvals and histories.
- Support multiple machine categories; the solution shall not be limited to packaging, gearbox or any single industry.

## 1.2 Scope included

- Organization, identity, roles, customer, supplier and item master data.
- Sales, machine-project initiation, project planning, engineering control and document revisions.
- BOM, material planning, warehouse, inventory, purchasing and supplier performance.
- Production, assembly, quality, NCR/CAPA, FAT, delivery, SAT and commissioning.
- Operational finance, project costing, workforce records, HR/payroll integration and after-sales service.
- Workflow approvals, alerts, dashboards, reports, audit trail, backup, recovery and external integrations.

## 1.3 Product rules

| Principle                      | Requirement                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------|
| One source of truth            | A transaction is created once and referenced downstream; duplicate re-entry is avoided.                                |
| Project traceability           | Every relevant commercial, material, engineering, production, quality and service transaction carries project context. |
| Controlled change              | Approved documents, BOMs, orders and test records are revised through history-preserving change control.               |
| Progressive integration        | Modules integrate when introduced; integration is not postponed until the final release.                               |
| Configuration over hard-coding | Taxes, numbering, workflow levels, departments, document types and statuses are configurable.                          |

## 2. Users and Organization Model

### 2.1 Organization hierarchy

**Organization → Branch → Physical Location → Department → Team / User**

- A company may operate multiple branches and physical locations.
- Departments may include Sales, Project Management, Mechanical Design, Electrical Design, Controls/Software, Purchase, Stores, Production, Quality, Finance, HR and Service.
- A user may belong to a department, hold one or more roles and be assigned to multiple projects.
- Authorization must support organization, location, department, project and record-level restrictions where required.

### 2.2 Baseline personas

| Persona              | Primary responsibilities                                                                |
|----------------------|-----------------------------------------------------------------------------------------|
| System administrator | Organization setup, users, roles, configuration, backup and integration administration. |
| Management           | Portfolio, finance, delivery, risk, capacity and profitability visibility.              |
| Sales                | Enquiry, quotation, revisions, customer communication and order conversion.             |
| Project manager      | Scope, milestones, work packages, risks, changes, cost and delivery control.            |
| Design engineer      | Requirements, tasks, drawings, BOM contribution, reviews and engineering changes.       |
| Purchase / Stores    | RFQ, comparison, PO, receipt, reservation, issue, stock and supplier follow-up.         |
| Production / Quality | Work orders, assembly progress, inspections, NCR/CAPA, FAT and release.                 |
| Finance / HR         | Invoices, receipts, payables, costing, employee, attendance, leave and timesheet data.  |
| Service              | Installed base, warranty, service calls, maintenance, spare parts and retrofit history. |

### 2.3 Access-control requirements

- Role-based permissions for view, create, edit, approve, cancel, export and administer actions.
- Separation of requestor and approver where configured.
- Inactive users cannot authenticate; ownership and audit history remain intact.
- Sensitive finance, payroll and personal records require restricted permissions.
- All privileged and approval actions are audit logged.

## 3. End-to-End Operating Flow

**Enquiry → Quotation → Sales Order → Machine Project → Engineering → BOM → Availability Check → Purchase → Assembly → Quality → FAT → Delivery → SAT / Commissioning → Invoice → Warranty → Service**

| Lifecycle stage    | Minimum controlled outcome                                                                |
|--------------------|-------------------------------------------------------------------------------------------|
| Commercial         | Capture requirement; issue controlled quotation; convert accepted order.                  |
| Project initiation | Create project, scope, owner, dates, commercial milestones and customer site.             |
| Engineering        | Allocate work packages; control requirements, drawings, reviews and revisions.            |
| Material planning  | Build progressive BOM; check stock; reserve available items; purchase shortages.          |
| Execution          | Issue materials; assemble; record effort, progress, tests, deviations and rework.         |
| Acceptance         | Run FAT; close punch items; obtain approval; deliver; install; complete SAT and training. |
| Commercial closure | Invoice by agreed milestones; record receipts, costs, margin and closure.                 |
| Lifecycle service  | Manage installed base, warranty, maintenance, breakdowns, spares and upgrades.            |

### 3.1 Mandatory traceability

- The system shall navigate from a machine project to its enquiry, quotation, order, requirements, tasks, documents, BOM revisions, requests, orders, stock, work orders, inspections, acceptance records, invoices and service history.
- Every downstream record shall show source references and current status without relying on free-text matching.
- Record cancellation shall not delete history; it shall record actor, timestamp, reason and replacement reference where applicable.

### 3.2 Stage gates

| Gate                      | Exit condition                                                          |
|---------------------------|-------------------------------------------------------------------------|
| G1 - Order accepted       | Approved sales order and project baseline exist.                        |
| G2 - Engineering released | Required drawings/BOM sections approved for procurement or manufacture. |
| G3 - Material ready       | Critical material is available or delivery risk is accepted.            |
| G4 - Assembly complete    | Production output and required inspections complete.                    |
| G5 - FAT accepted         | FAT approved and blocking punch items closed.                           |
| G6 - Site accepted        | SAT/commissioning, handover and training records complete.              |
| G7 - Project closed       | Commercial, material, documentation and open-action closure confirmed.  |

## 4. Core Data and Status Model

### 4.1 Principal records

| Domain             | Principal records                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Foundation         | organization, branch, location, department, user, role, permission, customer, supplier, item, UOM, tax, currency, warehouse           |
| Commercial         | enquiry, quotation, quotation revision, sales order, payment milestone, delivery note, sales invoice, customer payment                |
| Project            | machine project, machine, requirement, milestone, work package, task, risk, issue, change request, document, revision                 |
| Engineering        | BOM header, BOM revision, BOM line, alternative, manufacturer part, design review, release record                                     |
| Supply chain       | purchase request, RFQ, supplier quotation, comparison, purchase order, GRN, purchase return, stock transaction, reservation           |
| Execution          | production plan, work order, material issue, assembly operation, output, inspection plan, inspection, NCR, CAPA, rework               |
| Acceptance/service | FAT, punch item, shipment, installation task, SAT, commissioning, training, handover, warranty, service request, maintenance          |
| Business control   | approval, workflow step, comment, notification, audit event, invoice, payment, journal export, employee, attendance, leave, timesheet |

### 4.2 Standard record behavior

- All transactional records have a system ID, human-readable number, organization, creator, creation time, current owner and status.
- Project-relevant records include project and machine references.
- Documents support attachments, version, revision, approval status and superseded-by reference.
- Monetary records store currency, tax basis, net, tax and gross values with controlled rounding.
- Dates distinguish planned, promised, required, actual and closed dates as applicable.

### 4.3 Standard status model

| Status              | Meaning                                                 |
|---------------------|---------------------------------------------------------|
| Draft               | Editable; not operationally committed.                  |
| Submitted           | Sent for review/approval.                               |
| Approved / Released | Authorized for the next operational step.               |
| Rejected            | Not accepted; reason required.                          |
| Returned            | Sent back for correction without losing review history. |
| In progress         | Execution has started.                                  |
| Partially completed | Only part of quantity, value or scope is complete.      |
| Completed / Closed  | Exit conditions satisfied.                              |
| On hold             | Temporarily stopped; reason and owner required.         |
| Cancelled           | Stopped without deletion; reason required.              |

## 5. Release Plan Overview

| Release | Name                                  | Business outcome                                             |
|---------|---------------------------------------|--------------------------------------------------------------|
| R1      | Foundation & Master Data              | Trusted organization, identity and reusable masters          |
| R2      | Sales & Machine Project Initiation    | Accepted demand becomes a controlled project                 |
| R3      | Project Planning & Engineering        | Engineering responsibilities, dates and revisions controlled |
| R4      | BOM & Material Planning               | Progressive BOM becomes planned demand                       |
| R5      | Inventory & Warehousing               | Available, reserved and moving stock is visible              |
| R6      | Purchase & Supplier Management        | Shortages procured and supplier commitments tracked          |
| R7      | Production, Assembly & Quality        | Machine built with controlled inspection and rework          |
| R8      | FAT, Delivery & Commissioning         | Machine accepted, delivered and handed over                  |
| R9      | Finance, Workforce & After-Sales      | Cost, cash, people and installed-base lifecycle connected    |
| R10     | Enterprise Integration & Intelligence | Configurable, secure, integrated production platform         |

### 5.1 Release governance

- Each release includes analysis confirmation, UX, implementation, migration where needed, automated and manual testing, documentation and user acceptance.
- Foundational security, permissions, validation, auditability and backup are applied from Release 1 and expanded continuously.
- Basic module-specific approvals ship with the module; Release 10 adds a configurable enterprise workflow engine.
- A release is complete only when its acceptance criteria pass and no unresolved critical defect remains.

## 6. Release 1 - Foundation and Master Data

Establish controlled company structure, secure access and reusable business masters.

Organization → Users & Access → Customer / Supplier / Item Masters → Operational Readiness

### Functional scope

- Organization profile: legal and trading names, registration, tax/VAT/GST data, addresses, contacts, currency, timezone and fiscal settings.
- Operating branches, physical locations and departments; active/inactive control and responsible manager.
- Users, employees, roles, permissions, login credentials and access-control assignments.
- Customer master: customer code, names, billing/shipping/site addresses, contacts, tax data, currency, credit/payment terms and status.
- Supplier master: vendor code, addresses, contacts, tax data, categories, currencies, lead time, payment terms and status.
- Item master: code, description, type, category, UOM, make/buy, cost, selling price, manufacturer/part number, stock flag, reorder level and status.
- Warehouse master, tax/VAT configuration, currency, numbering series, document types and configurable reference statuses.

### Statuses

Draft, Active, Inactive, Blocked (where applicable).

### Reports and operational visibility

Master completeness, active users/roles, customer list, supplier list, item catalogue and change history.

### Release acceptance criteria

- Authorized administrators can configure organization hierarchy and access without code changes.
- Duplicate codes and invalid mandatory data are rejected.
- Role restrictions are enforced and privileged changes appear in the audit trail.
- Masters can be selected consistently by downstream modules.

RELEASE OUTCOME

**Establish controlled company structure, secure access and reusable business masters.**

# 7. Release 2 - Sales and Machine Project Initiation

Convert customer demand into an approved sales order and a fully traceable machine project.

Customer Enquiry → Quotation → Revision / Approval → Sales Order → Machine Project

## Functional scope

- Customer enquiry with machine/application summary, quantity, target delivery, customer site, source and attachments.
- Requirement summary covering intended use, capacity/performance, interfaces, utilities, constraints, standards and customer-provided items.
- Technical-commercial quotation with scope, exclusions, price, tax, validity, delivery, warranty and payment milestones.
- Quotation revisions, comparison, internal approval, customer acceptance/rejection and conversion history.
- Sales order with customer PO reference, ordered scope, value, delivery, payment milestones and special terms.
- Machine project auto-creation or controlled creation from an accepted order; project code, project owner, machine category, site, dates, value and baseline.
- Commercial attachments, customer communications, comments, activity timeline and status dashboard.

## Statuses

Enquiry: New/Qualified/Quoted/Won/Lost. Quotation: Draft/Review/Approved/Sent/Accepted/Rejected/Expired/Superseded. Order: Draft/Confirmed/On Hold/Partially Delivered/Completed/Cancelled. Project: Initiated/Active/On Hold/Closed/Cancelled.

## Reports and operational visibility

Sales pipeline, quotation conversion, quotations by customer, orders by month/customer, pending orders, payment milestone schedule and project intake.

## Release acceptance criteria

- An accepted quotation can create a sales order without re-entering lines and commercial terms.
- An approved order can create a machine project with source references.
- Every revision remains visible and only the current approved version is operational.
- Required project, customer, site, date and commercial values are validated.

RELEASE OUTCOME

Convert customer demand into an approved sales order and a fully traceable machine project.

## 8. Release 3 - Project Planning and Engineering

Plan and control multidisciplinary engineering work, decisions, documents, risks and changes.

Project Baseline → Milestones → Work Packages → Tasks → Design Review → Engineering Release

### Functional scope

- Configurable project template with phases, milestones, gates, planned dates and responsible departments.
- Mechanical, electrical, controls/software, documentation and other configurable work packages.
- Engineer/team allocation, task owner, dependencies, planned effort, dates, priority and completion evidence.
- Requirement register with source, owner, priority, verification method, status and links to design/test evidence.
- Document and drawing register with document number, type, revision, owner, review, approval and superseded history.
- Design review actions, comments, decisions and release approval.
- Risk, issue and action registers with probability/impact, mitigation, owner, due date and escalation.
- Engineering change request with reason, affected BOM/documents, cost/schedule impact, approval and implementation status.
- Timeline, progress, overdue work and project health dashboard.

### Statuses

Task: Planned/In Progress/Blocked/Review/Completed/Cancelled. Document: Draft/In Review/Approved/Released/Superseded/Obsolete. Change: Raised/Impact Review/Approved/Rejected/Implemented/Verified/Closed.

### Reports and operational visibility

Milestone performance, overdue tasks, workload by engineer/department, document status, requirement coverage, risk exposure and change impact.

### Release acceptance criteria

- Project managers can create and assign dependent work packages and tasks.
- Released drawings/documents cannot be overwritten; a new revision is required.
- Changes identify affected records and preserve approval history.
- Project progress is derived from controlled work, not only manual percentage entry.

RELEASE OUTCOME

Plan and control multidisciplinary engineering work, decisions, documents, risks and changes.

# 9. Release 4 - BOM and Material Planning

Transform evolving engineering design into controlled material demand before the full assembly is complete.

Preliminary BOM → Progressive Engineering BOM → Review → Release → Availability / Shortage Plan

## Functional scope

- Preliminary, engineering and manufacturing BOM types; multi-level assemblies and configurable BOM sections.
- BOM ownership by engineer/work package so multiple designers can contribute independently.
- Line attributes: item, description, quantity, UOM, reference, make/buy, required date, preferred supplier, manufacturer part, cost and notes.
- Create new items during BOM authoring subject to master-data validation/approval.
- Alternatives/substitutions, customer-supplied and non-stock items, scrap allowance and long-lead identification.
- Revision, comparison, line-level change history, approval and partial section release for procurement.
- Material availability, existing reservations, on-order quantity, shortage and required-by-date calculation.
- Estimated material cost versus project budget and released demand handoff.
- CSV/Excel import/export and preparation for CAD/SolidWorks BOM integration.

## Statuses

Working/Draft/In Review/Partially Released/Released/Revision Required/Superseded/Obsolete.

## Reports and operational visibility

BOM cost, revision variance, shortages, long-lead items, make/buy, unreleased sections and material readiness.

## Release acceptance criteria

- Multiple designers can contribute without waiting for the full assembly BOM.
- Only released quantities drive reservation or procurement demand.
- Every revision shows added, removed and changed lines.
- Demand is linked to project, machine, BOM revision and required date.

RELEASE OUTCOME

Transform evolving engineering design into controlled material demand before the full assembly is complete.

# 10. Release 5 - Inventory and Warehousing

Provide reliable stock visibility and project-specific material control.

Receipt / Stock In → Put-away → Available Stock → Reservation → Issue / Transfer / Return / Adjustment → Balance

## Functional scope

- Warehouse, zone/bin or configurable storage-location structure.
- Stock in, stock out, transfer, adjustment, cycle count and opening balance.
- Stock ledger and current balance by item, warehouse/location, batch/lot and serial number where enabled.
- Project reservation, allocation, deallocation, material issue and unused-material return.
- Available-to-promise calculation separating on-hand, reserved, available, inspection and blocked stock.
- Reorder/minimum stock, reorder proposals and low-stock alert.
- Negative-stock prevention by policy; reason and approval for controlled adjustments.
- Stock valuation method configurable at organization level and complete movement history.

## Statuses

Transaction: Draft/Posted/Reversed/Cancelled. Stock condition: Inspection/Available/Reserved/Issued/Blocked/Rejected.

## Reports and operational visibility

Current stock, project reservation, stock movement, valuation, ABC analysis, slow-moving items, shortages, cycle-count variance and reorder list.

## Release acceptance criteria

- Posted movements update balances atomically and cannot be edited directly.
- Reversal creates a linked counter-transaction and audit evidence.
- Users can see available, reserved, inspection and blocked quantities separately.
- Project material issue cannot exceed permitted available/reserved quantity.

RELEASE OUTCOME

Provide reliable stock visibility and project-specific material control.

# 11. Release 6 - Purchase and Supplier Management

Source project shortages, compare suppliers and control delivery, receipt and commercial status.

Purchase Request → Approval → RFQ → Supplier Quotation → Comparison → PO → GRN → Supplier Invoice → Payment Status

## Functional scope

- Purchase requests created manually or from released BOM shortages; requester, project, need date and justification.
- RFQ to one or more suppliers with issue date, response due date, terms and attachments.
- Supplier quotation capture for price, tax, lead time, validity, freight, payment terms, deviations and documents.
- Technical/commercial comparison and supplier selection with justification.
- Purchase order with project/BOM references, delivery schedule, terms, approval and supplier dispatch status.
- PO amendment, revision and complete history; closure of remaining quantity with reason.
- Partial receipt, GRN, rejected/accepted/inspection quantities, purchase return and pending balance.
- Supplier invoice matching record, payment status and interface-ready AP handoff.
- Supplier performance based on delivery, quality, response and price history.

## Statuses

PR: Draft/Submitted/Approved/Rejected/Converted/Closed. RFQ: Draft/Sent/Responses Received/Closed/Cancelled. PO: Draft/Pending Approval/Approved/Sent/Partially Received/Received/Closed/Cancelled. GRN: Draft/Posted/Inspection/Accepted/Rejected/Returned.

## Reports and operational visibility

Open POs, pending/delayed deliveries, purchase history, price history, supplier performance, GRN/inspection pending and project purchase cost.

## Release acceptance criteria

- Released BOM shortages can generate requests without duplicate manual entry.
- Supplier comparison records the selected offer and reason.
- Partial receipts update pending quantity and inspection stock correctly.
- PO amendments preserve the previous approved version and require reapproval by policy.

RELEASE OUTCOME

Source project shortages, compare suppliers and control delivery, receipt and commercial status.

# 12. Release 7 - Production, Assembly and Quality

Build the machine through controlled work orders, material consumption and verifiable quality.

Production Plan → Work Order → Material Issue → Assembly → Inspection / NCR → Output → Machine Complete

## Functional scope

- Production planning by project, machine, work centre/team, planned start/end and priority.
- Work orders for mechanical, electrical, controls/software and configurable manufacturing/assembly operations.
- Material issue against work order, actual consumption, return, substitution and variance.
- Operation/task progress, responsible person/team, planned/actual effort, blockers and completion evidence.
- Production output and finished-machine record with serial number and as-built BOM reference.
- Reusable inspection plans, characteristics, specification/tolerance, result, evidence and disposition.
- Incoming, in-process and final inspections; GRN, work order and finished-product triggers.
- NCR, containment, root cause, corrective action, preventive action, verification and closure.
- Rework and deviation/concession approval; quality history linked to item, supplier, work order and project.

## Statuses

Work order: Planned/Released/In Progress/Blocked/Quality Hold/Completed/Closed/Cancelled. Inspection: Planned/In Progress/Pass/Fail/Conditional Acceptance. NCR/CAPA: Open/Containment/Analysis/Action/Verification/Closed.

## Reports and operational visibility

Work-order status, WIP, material consumption/variance, productivity, inspection results, NCR ageing, rework and supplier-quality trend.

## Release acceptance criteria

- Work cannot be completed while mandatory inspections or blocking NCRs remain open.
- Actual consumption and output update project cost and inventory.
- Every failed characteristic has a controlled disposition.
- The finished machine retains serial, project, as-built BOM and quality traceability.

RELEASE OUTCOME

Build the machine through controlled work orders, material consumption and verifiable quality.

# 13. Release 8 - FAT, Delivery and Commissioning

Control customer acceptance, shipment, site execution and formal machine handover.

FAT Plan → Test → Punch List → Customer Sign-off → Dispatch → Installation → SAT → Commissioning → Handover

## Functional scope

- FAT plan, protocol/checklist, prerequisites, participants, scheduled date and required evidence.
- Test results, deviations, customer observations, punch items, owners, due dates and severity.
- FAT approval/sign-off and prevention of dispatch while blocking conditions remain.
- Packing list, delivery note, shipment/dispatch record, carrier details and delivery confirmation.
- Installation plan, site readiness checklist, assigned engineers, travel/site dates and activities.
- SAT and commissioning checklist, parameters, results, issues and customer acceptance.
- Customer operator/maintenance training records and attendance.
- Handover package: manuals, drawings, certificates, backup/configuration, spare list and signed records.
- Warranty start/end, open-item transfer and project closure checklist.

## Statuses

FAT/SAT: Planned/Ready/In Progress/Pass/Conditional Pass/Fail/Signed. Punch: Open/In Progress/Ready for Verification/Closed/Waived. Delivery: Planned/Packed/Dispatched/Delivered. Commissioning: Planned/In Progress/On Hold/Accepted/Closed.

## Reports and operational visibility

Upcoming FAT/SAT, open punch items, dispatch readiness, site schedule, commissioning status, handover completeness and delayed acceptance.

## Release acceptance criteria

- Dispatch readiness reflects FAT, quality, documentation and commercial controls.
- Customer sign-offs and evidence are stored against the project and machine.
- Blocking punch items prevent configured gate closure.
- Project closure requires approved handover and accountable transfer of open warranty items.

RELEASE OUTCOME

Control customer acceptance, shipment, site execution and formal machine handover.

# 14. Release 9 - Finance, Workforce and After-Sales

Connect money, people and the installed machine lifecycle to operational execution.

**Milestone / Cost Event → Invoice / Payable → Receipt / Payment → Project Margin | Handover  
→ Warranty → Service / Maintenance**

## Functional scope

- Accounts receivable/payable operational registers; customer and supplier invoices, due dates, allocation and partial payments.
- Cash, bank, bank-reconciliation support, account heads, journal entries, general ledger and fixed-asset register where full finance scope is approved.
- Trial balance, P&L, balance sheet, receivables ageing and payables ageing.
- Project cost aggregation for materials, purchasing, labour, production, rework, travel and configurable overhead; budget versus actual and margin.
- Employee master, department, designation, reporting manager, attendance, leave, overtime, timesheets and project allocation.
- Payroll inputs and payroll processing only for approved country/localization scope; otherwise secure export/integration to payroll provider.
- Installed-machine register with customer site, model/configuration, serial number, commissioning and warranty dates.
- Service request, breakdown, response, diagnosis, work performed, parts, time, travel, customer sign-off and closure.
- Preventive maintenance, maintenance-due alerts, service history, spare-part requests, retrofit and upgrade records.

## Statuses

Invoice: Draft/Issued/Partially Paid/Paid/Overdue/Cancelled. Service: New/Assigned/Scheduled/In Progress/Awaiting Parts/Resolved/Closed. Warranty: Pending/Active/Expired/Extended/Voided.

## Reports and operational visibility

Cash/receivables/payables, financial statements where enabled, project profitability, workforce utilization, attendance/leave/overtime, installed base, warranty, service SLA and maintenance due.

## Release acceptance criteria

- Commercial milestones can generate invoice requests and payment status feeds project visibility.
- Project costs reconcile to source transactions.
- Finance/payroll permissions protect sensitive information.
- Each delivered serialised machine has a complete warranty and service timeline.

RELEASE OUTCOME

**Connect money, people and the installed machine lifecycle to operational execution.**

# 15. Release 10 - Enterprise Integration and Intelligence

Complete configurable workflow, cross-module intelligence, external integration and production readiness.

Connected Data → Configurable Approvals → Notifications → Dashboards → Integration → Operational Intelligence

## Functional scope

- Workflow configuration, approval levels/matrix and multi-level approvals by role, user, department, amount, organization and document type.
- Approve, reject, return for correction, delegate, escalate, comment and full approval history.
- Email, dashboard and mobile/web push notifications; approved optional WhatsApp integration.
- Cross-module dashboards, drill-down reports, saved filters, exports and scheduled distribution.
- API, webhooks and integration monitoring for accounting, payroll, CAD/BOM, email and other approved systems.
- Customer and supplier portals subject to approved access and data exposure rules.
- AI-assisted delay, shortage, cost and risk indicators presented as recommendations with source evidence and user confirmation.
- Data validation, reconciliation, migration, backup/recovery validation, performance, security hardening and observability.
- End-to-end system testing, role testing, integration testing, recovery test, user acceptance and production cutover readiness.

## Statuses

Workflow: Draft/Active/Inactive. Notification: Queued/Sent/Delivered/Failed/Acknowledged. Integration: Active/Degraded/Failed/Disabled.

## Reports and operational visibility

Executive portfolio, project health, delivery, cost/margin, procurement risk, quality trend, supplier performance, utilization, service and audit/compliance.

## Release acceptance criteria

- Configured approvals enforce the expected sequence and authority limits.
- Users can drill from dashboard indicators to source records.
- Failed notifications/integrations are visible and safely retryable.
- Backup restoration, performance, security and end-to-end acceptance tests pass before production approval.

RELEASE OUTCOME

Complete configurable workflow, cross-module intelligence, external integration and production readiness.

## 16. Cross-Module Functional Requirements

| Capability            | System requirement                                                                                                                    |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Search and navigation | Global search by codes, names, project, machine serial, customer PO, supplier PO, item and document. Search results obey permissions. |
| Attachments           | Version-aware attachments with type, owner, timestamp and source record; controlled preview/download permissions.                     |
| Comments and activity | Record-level comments, mentions and chronological activity without replacing formal approvals or required fields.                     |
| Numbering             | Configurable unique numbering by organization/document/year; issued identifiers are not silently reused.                              |
| Approvals             | Basic approval at module launch; configurable matrix later. Self-approval is preventable by policy.                                   |
| Partial processing    | Purchase receipt, delivery, invoice and payment support partial quantities/values with outstanding balance.                           |
| Amendment             | Approved commercial and supply documents change through revision/amendment, never direct overwrite.                                   |
| Cancellation          | Reason, actor, date and downstream-impact check required. Posted/consumed records use reversal or closure rules.                      |
| Notifications         | Event-driven, role-aware, deduplicated and linked directly to the actionable record.                                                  |
| Export/print          | Permission-controlled PDF/CSV/XLSX output with organization identity, document number, revision and issue status.                     |

### 16.1 Core validation rules

- Required master and project references must be valid and active at transaction time.
- Quantities, currencies, UOM conversions, taxes and totals must use defined precision and rounding.
- Closed, cancelled, superseded and obsolete records cannot drive new operational transactions unless explicitly reopened by authority.
- A downstream transaction cannot exceed approved source quantity/value without approved tolerance or amendment.
- Status transitions are server-enforced and role-authorized; UI hiding alone is insufficient.

# 17. Security, Audit and Non-Functional Requirements

| Area                | Requirement                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Authentication      | Secure password storage, session expiry, account lock/rate limiting and optional MFA/SSO readiness.                             |
| Authorization       | Server-side role/permission checks; least privilege; restricted finance, payroll, personal and administrative access.           |
| Transport & storage | TLS for network traffic; database, backups and sensitive secrets protected according to deployment architecture.                |
| Audit trail         | Append-only history for login, create, edit, approval, status, permission, configuration, export and integration actions.       |
| Data integrity      | Database constraints, transactional posting, referential integrity, concurrency control and idempotent integration behavior.    |
| Availability        | Health monitoring, failure logging, restart procedures and agreed recovery objectives before production.                        |
| Backup/recovery     | Scheduled encrypted backup, retention, access control, restore testing and documented recovery procedure.                       |
| Performance         | List/filter/search and key transactions meet agreed response targets at validated concurrent-user and data volumes.             |
| Usability           | Responsive browser UI, consistent statuses/actions, accessible labels, clear errors, saved filters and minimal duplicate entry. |
| Localization        | Configurable timezone, date/number format, currency, language-ready labels and Tax/VAT/GST terminology.                         |
| Privacy             | Purpose-limited personal data, retention controls, export/delete handling subject to legal and audit obligations.               |
| Observability       | Structured application, integration and security logs with correlation IDs and actionable failure visibility.                   |

## 17.1 Audit event minimum fields

- Timestamp and timezone; organization; user/service identity; action; entity type and ID; project reference where applicable; before/after values or change summary; source IP/session where appropriate; outcome; reason/comment; correlation ID.

## 17.2 Production gate

- No open critical security, data-loss, financial-posting or authorization defect.
- Backup and restore have been executed successfully using production-equivalent data volume.
- Role matrix, audit events and integration failure handling are verified.
- Operational owner, support path and incident escalation are documented.

## 18. Reports, Dashboards and Notifications

| Domain     | Minimum reports                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|
| Sales      | Sales by customer/month, quotation conversion, pending orders, milestone billing.                       |
| Project    | Portfolio health, milestone variance, overdue tasks, risks, issues, changes and document readiness.     |
| Purchase   | Open PO, pending/delayed delivery, purchase history, price variance and supplier performance.           |
| Inventory  | Current stock, movement, valuation, reservation, ABC analysis, slow-moving, shortage and reorder.       |
| Production | Work-order status, WIP, material consumption, productivity, variance and machine readiness.             |
| Quality    | Inspection status, failures, NCR/CAPA ageing, rework, supplier quality and FAT/SAT readiness.           |
| Finance    | Receivables/payables ageing, trial balance, P&L, balance sheet and project profitability where enabled. |
| HR         | Attendance, leave, overtime, timesheet, allocation and utilization.                                     |
| Service    | Installed base, warranty, open service, SLA, maintenance due, parts and recurring failures.             |

### 18.1 Dashboard behavior

- Dashboards are role-aware and filterable by organization, branch, department, project, customer, date and status as applicable.
- Every indicator shows its definition and drills to the supporting records.
- Overdue and risk indicators use explicit business rules; colours alone are not the only status signal.
- Exports retain applied filters, generated time and generating user.

### 18.2 Minimum notifications

| Trigger family | Events                                                                                        |
|----------------|-----------------------------------------------------------------------------------------------|
| Approval       | Submission, pending approval, rejection, return, escalation and delegation.                   |
| Schedule       | Upcoming/overdue milestone, task, required material, FAT/SAT and commissioning activity.      |
| Supply         | RFQ response due, PO confirmation, delayed delivery, partial receipt and stock below minimum. |
| Quality        | Inspection due/fail, NCR assignment, CAPA overdue and blocking punch item.                    |
| Commercial     | Quotation expiry, invoice due/overdue, payment received and credit/hold condition.            |
| Service        | New service request, assignment, SLA risk, warranty expiry and machine maintenance due.       |

# 19. Integration and Data Migration

| Integration               | Minimum control                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------|
| Accounting                | Customer/supplier, invoice, payment and journal exchange; mapping, reconciliation and error queue. |
| Payroll                   | Employee/time/leave/overtime export or API; payroll result import where approved.                  |
| CAD / SolidWorks          | BOM import with item matching, new-item review, revision and validation report.                    |
| Email                     | Outbound notifications and optional controlled inbound association.                                |
| Messaging                 | Optional WhatsApp only after approved provider, consent, retention and template policy.            |
| Identity                  | Optional SSO/identity provider integration for enterprise deployment.                              |
| Customer/supplier portals | Restricted, audited access to explicitly published records only.                                   |

## 19.1 API requirements

- Versioned and authenticated endpoints with authorization, validation, rate control and documented errors.
- Idempotency for retried write operations and stable external reference mapping.
- Integration log with timestamp, direction, source, target, status, payload reference, correlation ID and retry history.
- No silent failure; failed records enter an actionable queue without duplicating successful transactions.

## 19.2 Migration requirements

- Define source owners, data templates, mandatory fields, code mapping and cleansing rules before import.
- Load masters before open transactions and opening balances.
- Produce pre-import validation, rejected-record report, reconciliation totals and signed migration acceptance.
- Preserve legacy identifiers as reference fields where operationally required.

## 20. Scope Boundaries, Dependencies and Delivery Governance

### 20.1 Explicit scope decisions

| Decision             | Required confirmation                                                                                                          |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Finance depth        | Confirm whether MachineIQ is the statutory book of record or exchanges operational postings with an accounting product.        |
| Payroll localization | Confirm countries, statutory deductions, payslip/legal reporting and update responsibility before full payroll implementation. |
| Tax configuration    | Use configurable Tax/VAT/GST; confirm jurisdictions, rates, reverse charge and document rules.                                 |
| CAD integration      | Confirm CAD products, export formats, part-number conventions and BOM revision ownership.                                      |
| Notifications        | Confirm email/mobile/WhatsApp providers, sender identity, consent and message retention.                                       |
| Deployment           | Confirm cloud/on-premise topology, availability, recovery objectives, user volume and data residency.                          |

### 20.2 Not implied without separate approval

- Country-certified statutory accounting, tax filing or payroll compliance beyond the approved localization.
- Native CAD authoring, PLC programming, MES/SCADA control or real-time machine control.
- Automatic AI decisions that approve, post, purchase, pay or change engineering records without an authorized user.
- Unlimited custom reports, third-party integrations, historical cleansing or customer-specific workflow variants.

### 20.3 Definition of done for every release

| Gate       | Completion condition                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------|
| Scope      | Approved stories/workflows match this specification and documented release clarifications.                    |
| Function   | Happy path, exceptions, partial processing, permissions and status transitions pass.                          |
| Data       | Validation, constraints, audit, imports and reports reconcile.                                                |
| Quality    | Automated tests and manual regression pass; no critical defect remains.                                       |
| Security   | Authorization and sensitive-data checks pass; dependency/security findings are resolved or formally accepted. |
| Operations | Backup, monitoring, error handling, deployment and rollback are documented and tested as applicable.          |
| Acceptance | Demo completed, acceptance evidence retained and authorized business owner signs off.                         |

# A. Client Requirement Traceability Matrix

This appendix confirms that the original client document is retained while correcting sequence and adding the machine-project lifecycle.

| Original requirement                              | Covered in   | Treatment                                                                              |
|---------------------------------------------------|--------------|----------------------------------------------------------------------------------------|
| Company, branches, locations, departments         | R1           | Expanded organization hierarchy                                                        |
| Users, employee, role, permission, login          | R1 / R9      | Identity in R1; workforce depth in R9                                                  |
| Customer, supplier and item master                | R1           | All original fields retained and normalized                                            |
| Customer enquiry to payment receipt               | R2 / R8 / R9 | Commercial initiation, delivery and finance linked to project                          |
| Purchase request to supplier payment              | R6 / R9      | Includes RFQ, comparison, PO revisions, GRN and AP status                              |
| Warehouse, stock in/out/transfer/adjustment/count | R5           | Adds project reservation, traceability and valuation                                   |
| BOM to finished goods                             | R4 / R7      | Engineering BOM separated from execution to support progressive design                 |
| Incoming, in-process, final inspection            | R7           | Reusable inspection framework                                                          |
| NCR and CAPA                                      | R7           | Adds containment, root cause, verification and closure                                 |
| Employee, attendance, leave, overtime, payroll    | R9           | Payroll depth subject to localization decision                                         |
| AR, AP, GL, cash, bank, assets                    | R9           | Full finance depth subject to book-of-record decision                                  |
| Workflow master, steps and history                | R10          | Basic approvals earlier; advanced configurable engine in R10                           |
| Email, dashboard, mobile, WhatsApp                | R10          | Provider/policy-dependent channels                                                     |
| Sales reports                                     | R2 / R18     | Sales by customer/month and pending orders retained                                    |
| Purchase reports                                  | R6 / R18     | Supplier performance and open PO retained                                              |
| Inventory reports                                 | R5 / R18     | Valuation, ABC and slow-moving retained                                                |
| Production reports                                | R7 / R18     | Work status, consumption and productivity retained                                     |
| Finance reports                                   | R9 / R18     | Trial balance, P&L; and balance sheet retained where finance enabled                   |
| PostgreSQL recommendation                         | Architecture | Supported as primary relational database direction; final topology approved separately |
| Machine maintenance due alert                     | R9 / R18     | Backed by installed base and maintenance schedule                                      |

TRACEABILITY  
RESULT

**All substantive requirements from the original client document are represented. Invalid artefacts such as 'Plain Text', 'Show more lines' and the '3GL' typo are intentionally excluded or corrected.**

## B. Approval and Change Record

Approval of this document establishes the baseline release direction. Detailed UI designs and user stories may refine implementation without contradicting the workflows, controls or acceptance boundaries defined here.

| Role                    | Name | Signature | Date |
|-------------------------|------|-----------|------|
| Client sponsor          |      |           |      |
| Client process owner    |      |           |      |
| MachineIQ product owner |      |           |      |
| Technical lead          |      |           |      |

### Change record

| Version | Date        | Change summary                                                                                       | Approved by |
|---------|-------------|------------------------------------------------------------------------------------------------------|-------------|
| 1.0     | 07 Sep 2026 | Initial controlled specification consolidating client ERP scope and corrected OEM machine lifecycle. | Pending     |

BASELINE

**MachineIQ: one connected operational record from customer requirement to machine lifecycle.**